Title:

Top Movies & Series – Client-Side Movie Ranking Application

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1. Introduction

This project is a client-side web application designed to discover and rank movies and television series using data retrieved from The Movie Database (TMDB) API.

The primary objective was to create a dynamic ranking system that balances rating, vote count, and popularity rather than relying on a single metric.

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2. System Architecture

The application follows a client-side modular architecture:

- Presentation Layer (HTML + CSS)
- Logic Layer (JavaScript)
- External Data Layer (TMDB REST API)

Data is fetched asynchronously using `async/await`, normalized, scored, cached, and rendered dynamically.

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3. Data Processing & Ranking Model

To ensure balanced ranking, a custom scoring algorithm was implemented:

Score = (Rating × 2.5) + (log10(Votes) × 2) + (Popularity × 0.02)

This approach prevents high-rated but low-voted items from dominating the list and balances popularity influence.

The normalization layer converts raw API responses into structured internal objects.

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4. State Management

The application maintains client-side state including:

- Items dataset
- Favourites list (persisted via LocalStorage)
- Genre mapping
- Cached API responses
- Sorting direction

This ensures efficient re-rendering and reduces redundant network calls.

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5. Performance Optimizations

Several performance improvements were implemented:

- Caching detailed movie responses
- Debounced search input (250ms delay)
- Batched asynchronous requests
- Deduplication using ID mapping
- Lazy fallback handling for broken images

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6. User Interface Design

The UI follows a component-based design using:

- CSS custom properties
- Responsive grid layout
- Modal interaction system
- Animated transitions
- Toast notification system
- Adaptive breakpoints

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7. Conclusion

This project demonstrates proficiency in:

- Asynchronous JavaScript
- REST API integration
- Client-side data processing
- UI/UX design

- State management
- Performance optimization